SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I Subash G [192572150] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

<u>RUBRICS FOR CALCULATION:</u>

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

SOLUTIONS

Question 1:

Problem Statement:

A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Types of Agents:

- ❖ Simple Reflex Agent
- ❖ Model-Based Agent
- ❖ Goal-Based Agent
- ❖ Utility-Based Agent

Definations of those Agents:

Simple Reflex Agent

A simple reflex agent makes decisions based only on the current situation. It follows predefined rules and does not consider past experiences.

For example, if motion is detected in a room, the system immediately turns on the lights. Similarly, if the temperature rises above a certain level, the air conditioner is switched on. These agents are fast and easy to implement, but they cannot learn user preferences or adapt to changing conditions.

Model-Based Agent

A model-based agent maintains information about the environment and uses previous observations to make decisions.

For example, if a room was occupied a few minutes ago, the system may keep the lights on even if no motion is currently detected. It can also remember temperature trends and adjust cooling accordingly. This makes the system more reliable than a simple reflex agent because it has knowledge about the current state of the environment.

Goal-Based Agent

A goal-based agent selects actions that help achieve a specific objective.

In a smart home, the goals may include maintaining a comfortable room temperature, reducing electricity consumption, and ensuring safety. If the temperature becomes too high, the agent decides whether to turn on the fan or air conditioner based on the goal of achieving comfort. It evaluates different actions before making a decision.

Utility-Based Agent

A utility-based agent chooses the action that provides the highest overall benefit. It considers multiple factors simultaneously and selects the best option.

For example, during a hot afternoon, the system may decide whether to use natural ventilation or air conditioning. It compares factors such as comfort, energy consumption, and cost before choosing the most beneficial action. This agent provides a balanced solution when multiple objectives are involved.

Most effective Approach:

A hybrid approach combining model-based, goal-based, and utility-based agents would be the most effective for a smart home system.

The model-based component helps the system understand the environment and user behaviour. The goal-based component ensures that comfort and security objectives are achieved. The utility-based component helps select the most efficient action by balancing comfort, energy savings, and security.

Therefore, a hybrid intelligent agent system is the best choice because it can learn user preferences, adapt to changing conditions, and make optimal decisions. This results in improved comfort, better energy efficiency, and enhanced home security.

QUESTION 2:

Problem Statement:

A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Uniform Cost Search (UCS) Defination:

Uniform Cost Search expands the node with the lowest path cost first. It always chooses the least expensive path available and continues exploring until the goal is found.

In a maze, if different paths have different movement costs, UCS helps the robot find the path with the minimum total cost. It guarantees an optimal solution when all costs are positive.

Advantages of UCS:

- Finds the optimal path.
- Suitable when path costs are different.
- Complete algorithm that always finds a solution if one exists.

Disadvantages of UCS

- Requires large memory to store explored nodes.
- Can be slow in large search spaces.
- Time and space complexity increase significantly as the number of states grows.

Iterative Deepening Search (IDS) Defination:

Iterative Deepening Search combines the advantages of Depth First Search and Breadth First Search. It repeatedly performs depth-limited searches with increasing depth limits until the goal is found.

In a maze, the robot first searches at depth 1, then depth 2, then depth 3, and so on until the exit is discovered.

Advantages of IDS:

- Requires very little memory.
- Finds the shallowest solution.
- Suitable when the depth of the goal is unknown.

Disadvantages of IDS:

- Repeats the same searches multiple times.
- May take longer because previously visited nodes are explored again.

Comparison of UCS and IDS:

Uniform Cost Search is better when path costs vary because it guarantees the least-cost path. However, it requires more memory. Iterative Deepening Search uses less memory and is suitable when the goal depth is unknown, but it may perform repeated searches.

Relation to Intelligent Agents:

A simple reflex agent would not perform well in an unknown maze because it only reacts to the current situation. A model-based agent is more suitable because it can remember previously visited locations and build knowledge about the maze. A goal-based agent is even more effective because its objective is to reach the exit. It evaluates different paths and selects actions that move it closer to the goal.

Most Effective Combination:

The most effective combination would be a **Goal-Based Agent with Uniform Cost Search.**

The goal-based agent keeps the robot focused on reaching the exit, while Uniform Cost Search ensures that the path chosen has the minimum cost. This combination improves decision-making and helps the robot find an efficient and optimal route through the maze.

Uninformed search algorithms play an important role in solving navigation problems. Among the available approaches, a Goal-Based Agent using Uniform Cost Search provides the best balance between intelligent decision-making and optimal path finding, making it highly suitable for robot navigation in an unknown maze.